

WORK EXPERIENCE

- **Citi** New York, NY
Software Engineer VP - Risk Analytics Products April 2025 - Present
Software Engineer AVP August 2023 - March 2025
Software Engineer Contract May 2022 - July 2023
 - Build Citi's new vectorized risk ratings engine for wholesale credit
 - Build an AI-enabled with human-in-the-loop ETL framework for risk reporting
 - Cut wholesale loss model runtimes by 90%+ using distributed graph computation
 - Build a compiler for a type-safe DSL with UDF support for faster and safer numerical computations
 - Build a realtime schemaless report generation framework with DuckDB and Parquet
- **OANDA** Toronto, ON
Software Developer Intern - Trading Engine / Execution May 2020 - August 2021
 - Optimize polling logic in multi-queue ZeroMQ cases to reduce CPU usage by 90%+ and improve responsiveness
 - Investigate memory fragmentation issues in TCMalloc for long running processes with high memory usage
 - Improve stability of reporting and notification services running on Twitter Finagle
 - Migrate a group of services out of a large monorepo to facilitate faster and more atomic releases

EDUCATION

- **University of Toronto** Toronto, ON
Bachelor's Degree - Computer Science Specialist and Applied Statistics Minor September 2018 - April 2023
 - Honours Bachelor of Science with High Distinction: 3.85/4.00 CGPA
 - University of Toronto Scholar – 2019, 2020, 2022, and 2023 Dean's List Scholar

INVOLVEMENTS

- **Imglabs.io** Toronto, ON
Co-Founder April 2022 - Present
 - Distributed image proxy written in Rust and deployed using WebAssembly to 300+ edge locations
 - Currently in production with commercial customers serving 10TB+ monthly
 - Industry-leading cost efficiency and performance: \$10 per million images with no egress or other costs
 - Memory allocator optimizations to keep memory usage below 128MB platform limit
 - Software stack: Rust, WebAssembly, JavaScript, React/Next.js, Cloudflare Workers, AWS S3, ClickHouse, Prometheus, Grafana, Supabase, Docker, Stripe

PROJECTS

- **Scaler** (finos/opengris-scaler): Efficient, lightweight and reliable distributed computation engine
- **Pargraph** (finos/opengris-pargraph): Drop-in Graph/DAG generation and computation library for Python
- **RAG Using Qdrant with Polymarket (demo)**: Fast search and summarization of real-time event contracts
- **Columnar Financial Statements Querying (demo)**: Query SEC Form 10-Qs in the browser using DuckDB WASM
- **Dsfs** (darenliang/dsfs): A highly-experimental filesystem in userspace using FUSE backed by Discord attachments
 - Append-only log-structured filesystem: Supports automatic snapshots and rollback
 - Realtime synchronization: Syncing files and folders between remote clients in realtime
- **Nikel API** (nikel-api/nikel): Free and open data API for the University of Toronto (discontinued)
- **MikuBot** (darenliang/MikuBot): General purpose Discord bot serving more than 9,000 servers (discontinued)

SKILLS

- **Languages**: Python, Go, C/C++, Rust, JavaScript/TypeScript, Java, Scala, SQL
- **Assessments**: CodeSignal: 600/600